

SPMM
Smart
Revise

Addiction Psychiatry I

Paper B

Syllabic content 11.1

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1. Epidemiology of substance use

A. Alcohol Use:

Epidemiological feature	Rates
Percentage of general population who drank alcohol in the last week in the UK	67% men; 53% women
Percentage of adults who drank above the recommended limits among those who drank alcohol in the last week	55% men (24% of general population); 53% women (18% of general population)
Percentage of school pupils (aged 11-15) who had drunk alcohol at least once	43%
Patients presenting to primary care that consume alcohol at harmful or hazardous levels	20%
Annual prevalence of hazardous drinking (AUDIT score ≥ 8) in UK households	38% of men; 15% of women; 27% of White adults, 18% of Black adults, 8% South Asian adults
Peak age of hazardous drinking	16 to 19 (women); 20-24 years (men)
Prevalence of alcohol dependence	74 per 1,000 overall; 119 per 1,000 men and 29 per 1,000 women.
Number of all hospital attendees/admissions that are alcohol related	1 in 16 hospital admissions (Pirmohamed, 2000); 1 in 6 A&E attendees (this rises to 8 in 10 during peak hours) (HEA, 1998).
Age-specific use patterns in alcohol dependent people	Age at first drink (13-15), age at first intoxication (15-17) and age at first 'problem' related to alcohol (16-22) is essentially the same as the general population for alcohol dependent patients. But the age at death is around 60 years – very premature.
Alcohol use during pregnancy	One in 10 pregnant women drank in the last week

B. Drug Use:

Epidemiological feature	Rates
Percentage of UK adults (16 to 59) who took an illicit drug in the last year	1 in 12 (8.3%); frequent users i.e. >once/month = 2.8%. (Cannabis (6.4%) followed by powder cocaine (1.9%) and ecstasy (1.3%))
Percentage of young adults (16 to 24) taking any drug in the last year in the UK	1 in 6 (16.3%); frequent users = 5.1%
Percentage of adults aged 16 – 59 had taken a Class A drug in the last year	2.6%
Percentage of school pupils who took an illicit drug in the last year in the UK	12% (raises to 17% if all past exposures are considered)
Percentage of drug users in the last year who use multiple substances	61% when alcohol is also included; 7% for other drugs (not alcohol)
The most commonly reported age for first taking drugs.	Cannabis - 16 years; powder cocaine and ecstasy - 18 years
Average duration of drug use among regular users.	Cannabis - 6.2 years; powder cocaine (4.4 years) or ecstasy (3.9 years)

The above data were obtained from Health & Social Care Information Centre. Statistics on Drug Misuse: England 2013 and Statistics on Alcohol: England 2014 (www.hscic.gov.uk; published 29th May 2014)

2. Typology & Classification

A. Legal categorization:

1971 MISUSE OF DRUGS ACT UK
Class A drugs
Include: Ecstasy, LSD, heroin, cocaine, crack, magic mushrooms (whether prepared or fresh), methylamphetamine (crystal meth), other amphetamines if prepared for injection
Class B drugs
Include: Amphetamines, Methylphenidate (Ritalin), Pholcodine.
Class C drugs
Include: Cannabis, tranquilisers, some painkillers, GHB (Gamma-hydroxybutyrate), ketamine

B. Psychiatric nosology:

- ★ Magnus Huss in 1865 coined '**alcoholism**' which included dependence, acute alcohol-related health issues and cumulative social problems.
- ★ ICD10 has a diagnostic code for '**harmful use**' wherein the actual damage is caused to the drinker physically or mentally, but he has no dependence pattern (yet).
- ★ **ICD-10 alcohol dependence** requires at least 3 out of following list satisfied in last 12 months:
 1. Intense desire to drink alcohol
 2. Difficulty in controlling the onset, termination and the level of drinking
 3. Experiencing withdrawal symptoms if alcohol is not taken
 4. Use of alcohol to relieve from withdrawal symptoms
 5. Tolerance as evidenced by the need to escalate dose over time to achieve same effect
 6. Salience – neglecting alternate forms of leisure or pleasure in life
 7. The narrowing personal repertoire of alcohol use.
- ★ DSM IV had a concept of '**abuse**' which refers to maladaptive use defined as below:
 1. Despite problems in social, occupational, physical and psychological domains
 2. In hazardous situations
 3. At least one month, recurring over a longer period usually.
 4. But not dependent on alcohol.
- ★ **DSM IV alcohol dependence** requires at least 3 out of following list lasting for at least a month:
 1. Consuming alcohol for longer period and in larger amounts than intended
 2. Unsuccessful attempts to cut down

3. Experiencing withdrawal symptoms if alcohol is not taken
4. Use of alcohol to relieve from withdrawal symptoms
5. Tolerance as evidenced by the need to escalate dose over time to achieve same effect (at least 50% increase from start)
6. Salience – most time of life spent on pursuing alcohol directly or indirectly
7. Failure in role obligations and physical health
8. Giving up alternate pleasures
9. Continued use despite knowing the harm caused

★ Major changes in DSM 5

- DSM-5 combines the DSM-IV categories of substance abuse and substance dependence into a single Substance Use Disorder (SUD) that is measured on a continuum from mild (replacing abuse) to severe (replacing dependence). The same broad criteria are used for the diagnosis of specific substance-related disorders.
- DSM-IV substance abuse required only one symptom for a diagnosis, but in DSM-5, mild substance use disorder requires two to three symptoms from a list of 11.
- While drug craving has been added to the list of diagnostic features, the description of legal problems has been removed to reduce cultural variations that preclude international comparisons.
- Furthermore, the topic of Addictive Disorders in DSM-5 is much broader, encompassing non-substance related addictions as well. The gambling disorder is the only condition currently included in the new category of behavioural addictions. Note that in DSM-IV pathological gambling was listed under impulse control disorders.
- Internet gaming disorder has been included in Section III (disorders requiring further research before formal diagnostic description). Similarly, caffeine use disorder is included in Section III of DSM-5.

C. The concept of dependence:

Dependence is not a moral, but a medical concept and its historical development predominantly centered on the use of alcohol though it is now extended to be used for different illicit drugs. It focuses on treatment and indicates intensity, and predicts the outcome to some extent. It also indicates the need for abstinence compared to reduced drinking in some users.

Edwards & Gross criteria for alcohol dependence (1976) included the following criteria:

1. Narrowed repertoire
2. Salience of alcohol-seeking behaviour
3. Increased tolerance
4. Repeated withdrawals
5. Drinking to prevent or relieve withdrawals.

6. Subjective awareness of compulsion
7. Reinstatement after abstinence

Neuroadaptation is a term used to differentiate a certain state from dependence. In case of SSRIs, for example, discontinuation, loss of effect, relief from discontinuation on taking the same drug, etc. can be present but no active drug seeking behaviour develops. This is called neuroadaptation. Remarkably this starts developing as soon as the first dose is taken.

- ★ **Narrowing of repertoire:** A social drinker may drink in various situations at various places and may have various drinks of his choice. A dependent drinker will settle into a fixed setting, sticking to one beverage. This may help him maintain a steady level of alcohol in the blood without much fluctuation. Note that some widening of repertoire is inevitable in the career of an alcohol drinker – moving from weekends only to all days of the week, etc.
- ★ **Salience** refers to the neglect of all leisure and alternate forms of pleasure apart from alcohol. One's life revolves around getting alcohol, storing it, saving money to buy it, making opportunities to drink, etc.
- ★ **Tolerance:**
 - It has behavioural, pharmacokinetic and pharmacodynamic cellular mechanisms.
 - Tolerance may be due to increased excitability of neurons when using depressants; in this case withdrawal reactions will inevitably co-occur. Tolerance may also occur when the potency of alcohol taken reduces due to other reasons such as pharmacokinetic changes. In this case, withdrawal may or may not be present.
 - Tolerance must be differentiated from withdrawal. Intermittent use is unlikely to lead to withdrawal.
 - Tolerance may occur even in intermittent use.
 - Note that regular use of benzodiazepines causes much withdrawal anxiety but not much tolerance is seen for the anxiolytic effects (when compared to hypnotic effects); this is in contrast to barbiturates.
 - 'Diminished tolerance' is also a notable phenomenon in drinkers. This may be due to
 - When drinking alcohol after a period of abstinence, the tolerance may revert to normal and might lead to quick intoxication.
 - Older patients with brain damage may have curtailed neuroadaptation leading to reversed tolerance.
 - Metabolic problems such as liver disease may apparently look like diminished tolerance.
 - Idiosyncratic in certain individuals

Withdrawal:

- This refers to physical and psychological symptoms due to non-availability of alcohol in a dependent user.

- The pharmacology of 'reward' is different from the pharmacology of withdrawal. Cocaine induces modest withdrawal features but has very high reward potential.
- A user in withdrawal gives a very important opportunity towards permanent behaviour change.

Compulsion refers to repetitive intense drug seeking with an urge. This is different from craving which is a motivational state occurring in withdrawal.

Craving is a motivational state where emotional valence can be positive or negative. Tiffany identified the following components of craving;

1. Urge/compulsion
2. Intention/plan – readiness to yield
3. Expectation of a satisfying outcome
4. Anticipation of decrease in pain / relief from negative effects
5. Loss of control

Some alcohol dependent patients may have a cognitive set characterized by the fear of craving.

Reinstatement on relapse: This is a criterion stressed by Edward & Gross in their early description of alcohol dependence. A patient, who is abstinent for a long time, may go back to the original pattern of heavy drinking in a surprisingly rapid manner after having a priming dose of alcohol (relapse). Various explanations for this include:

1. *Abstinence violation effect* – the cognition that 'I had a drink; so I am a drinker again' may force patients to reinstate full pattern of drinking
2. Propensity to experience withdrawal symptoms may be carried over even through the period of abstinence in some individuals
3. Cues may trigger memory after a priming dose, making compulsive drinking a behaviour.

Impaired control:

- Not in original Edwards & Gross criteria – but popularized in ICD and DSM.
 - This is explained in two different ways
1. Within a single episode of drinking, one loses his control on the intended amount of alcohol and ends up being often intoxicated.
 2. Considering one's overall 'alcohol career' it is evident that one tries to cut down repeatedly but fails every time.

Cloninger's classification:

Type 1	Type 2
Milieu limited	Male limited
Males and females	Males usually
Loss of control	Inability to abstain

No strong family history	Strong heritability
Not much criminality	Antisocial traits high
Starts >25years	Starts <25 years

Jellinek's classification:

<i>Type</i>	<i>Description</i>
Alpha α	Psychological dependence; undisciplined not progressive; no withdrawal; major problems are in interpersonal domain only
Beta β	Physical damage but no dependence
Gamma γ	Loss of control plus physical dependence. Withdrawal seen; earlier stages are similar to alpha – most common among AngloSaxons
Delta δ	No loss of control but unable to abstain e.g., in France this may be socially accepted – no disapproval or interpersonal problems.
Epsilon ϵ	Dipsomania – binges and bouts

3. Pharmacological and physiological effects

A. Alcohol:

- ★ **Pharmacology:** Intercalates into the fluid cell membrane; decreases NMDA sensitivity; increases GABA sensitivity; down-regulates calcium channels; up-regulates nicotine receptor gated sodium channels.
- ★ Females have lower body water, a higher alcohol absorption rate and lesser alcohol dehydrogenase in their gut mucosa compared to males. All of these lead to a higher oral availability in females when compared to males.
- ★ **Features of alcohol withdrawal:** Usually starts within 12 hours of last drink. Tremor, diaphoresis, sleeplessness, GI distress and anxiety are prominent. Increased urge and craving for alcohol may be seen. The severity of symptoms depends on the degree of pre-existent drinking. If unattended symptoms may peak in 48 hours. In these cases, seizures may occur. These are grand-mal seizures especially common in those who had previous seizures (epilepsy or withdrawal seizures), head injury and electrolyte imbalance such as hyponatraemia or hypokalemia. In around 5% patients, **delirium tremens** may set in. Disturbed autonomic functions (pulse, temperature and blood pressure changes in either direction), clouded consciousness with hallucinations (often Lilliputian) and agitation can occur.
- ★ **Seizures:** The incidence of seizures quoted as 8% in untreated alcohol-dependent patients. This risk drops to 3% if withdrawal is being treated. This may be due to a kindling process whereby episodic alcohol withdrawal sensitises the brain leading to increased likelihood of a seizure with each future episode (true for all withdrawal symptoms – not just seizures).
- ★ **Delirium Tremens:** 30% (not all) of the patients with withdrawal seizures later develop DTs. Risk factors for DT and seizures include severe dependence, past history of DTs, being an older patient and having an acute physical illness. 10% of those with DT may die if the withdrawal is untreated. In patients experiencing polysubstance withdrawal that includes alcohol, one must treat alcohol withdrawal DTs first, due to the significant mortality risk.

Time course of alcohol withdrawal

- Onset of shakes (4-12 hours)
- Onset of perceptual disturbances (8-12 hours)
- Seizure onset (12-24 hours); peaks at 48 hours
- Delirium onset (72 hours)

Opioids:

- ★ The term **opiate** is often used interchangeably with an **opioid**, but 'opiate' refers only to natural opium alkaloids and the semi-synthetics derived from the alkaloids. Opioids have potent analgesic, euphoriant and anxiolytic effects for which they are abused.
- ★ **Heroin** is the most commonly used opioid and it is most commonly smoked (called **chasing**) or injected IV/IM though it can be snorted or taken orally too. Street heroin is almost always adulterated and impure; accidental injection of the pure form may result in death due to toxicity.
- ★ Once only use of heroin is quite rare – generally people progress from recreational exposure to dependent pattern mostly. The current *prevalence* of heroin use in the UK is around 1% (male-to-female ratio of 2:1), with most treatment seekers being in their 20s.
- ★ 3 receptors are important in opioid physiology. Mu (μ), kappa (κ) and delta (δ) – all three are G protein-coupled. Most abused opioids are Mu receptor agonists (morphine-like). This mediates analgesia, respiratory depression, euphoria (only in some), drowsiness and constipation. They increase dopamine release in VTA. Mu agonists induce tolerance for most actions, but little or no tolerance is noted for constipation and miosis. Some cross-tolerance is seen among different mu agonists. Kappa agonists produce dysphoria, no pupillary effect but some analgesia. They decrease dopamine release in VTA (Kaplan & Sadock CTP, 2010).

Drug	Routes	Half life	Strength
Morphine	Multiple routes of intake; low oral bioavailability (30%)	2-3 hours (both oral and iv)	Parenteral use 3 times stronger than oral
Heroin (diamorphine)	Multiple routes of intake; poor oral bioavailability (30%)	<3 minutes	4-5 times stronger than oral morphine
Methadone	80% oral bioavailability	15 to 60 hours	3-4 times stronger than oral morphine
Buprenorphine	Sublingual, transdermal or injected	30h if taken sublingually (1-7h if taken orally)	40 times more potent at receptor level, but a partial agonist
Oxymorphone, hydromorphone	Oral	3 to 7 hours	5-7 times more potent
Codeine, dihydrocodeine, pethidine	Oral (Pethidine – intravenous)	2-4 hours	0.1 to 0.2 times potent compared to morphine

Adapted from Luty J. 2014 BJPpsych Advances

- ★ **Intoxication effects**
 - Initial euphoria
 - Apathy & dysphoria follows.

- Psychomotor agitation or retardation
- Pupillary constriction (if pupillary dilation is seen the overdose may be very severe and anoxia has set in)
- Drowsiness or coma
- Slurred speech
- Impairment in attention or memory

★ **Withdrawal symptoms** (*cold turkey*): Opioid withdrawal is rarely fatal in a healthy adult. Morphine and heroin withdrawal syndrome begins 6 to 8 hours after the last dose, peaks in 2 days and reduces in a week usually.

- Dysphoric mood
- Nausea or vomiting
- Muscle aches
- Lacrimation or rhinorrhea
- Pupillary dilation, piloerection (gooseflesh), or sweating
- Diarrhea
- Yawning
- Fever
- Insomnia

Note that insomnia, bradycardia, temperature dysregulation, and a craving for opioids can persist for months after an episode of withdrawal, necessitating maintenance methadone in number of patients.

Cannabis:

- ★ **Pharmacology:** Cannabis is obtained from the plant **Cannabis sativa**. The principal component of cannabis is **Δ 9-THC**. The cannabinoid receptor is an inhibitory **G protein** (Gi) linked receptor where endogenous cannabinoids called **anandamides** act. These receptors are found in highest concentrations in the basal ganglia, the hippocampus, and the cerebellum, and are sparse in the cerebral cortex and none in the brainstem; this may explain cannabis' minimal effects on respiratory and cardiac functions.
- ★ **Physiological effects:** The euphoric effects appear within minutes after smoking, peak in about 30 minutes, and last 2 to 4 hours. Tolerance to cannabis has been demonstrated in both human and animal subjects. Psychological dependence has also been found. But the evidence for physiological dependence is not strong.
- ★ **Withdrawal** symptoms in humans are irritability, insomnia, anorexia and mild nausea. These are very rare and occur only when very high doses are suddenly stopped.

- ★ **Intoxication** is characterised by impaired motor coordination, euphoria, a sensation of slowed time, social withdrawal, conjunctival injection, increased appetite (munchies), dry mouth, tachycardia. In some cases, cannabis intoxication is associated with heightened perceptual sensitivity, depersonalization and derealization. Impairment in **motor skills** may remain long after the euphoriant effects are gone.

Benzodiazepines (Sedatives/Hypnotics):

- ★ Abusers include those who were prescribed at some point, but became dependent and use supranormal doses obtained illegally, those who use benzodiazepines in combination with other drugs, e.g. to boost methadone or come off stimulants and those dependent on legitimate prescriptions, not able to stop them (non-abusers).
- ★ Rates of dependence varies with duration of use – 15% of those using 1-5 months of prescribed benzodiazepines develop dependence compared to nearly 40% of those using it for nearly a year (De lasCuevas, 2003)
- ★ **Symptoms of intoxication** includes slurred speech, incoordination, unsteady gait, nystagmus, impairment in attention or memory, stupor or coma, behavioural changes such as inappropriate sexual or aggressive behaviour, mood lability, and impaired judgment.
- ★ **Symptoms of withdrawal** includes prominent anxiety and autonomic hyperactivity, increased tremor, insomnia, nausea or vomiting, transient visual, tactile, or auditory hallucinations or illusions, psychomotor agitation, in some cases - grand mal seizures. Kinaesthetic hallucinations are also reported in some individuals. Note that withdrawal delirium can be fatal.
- ★ Using prescribed benzodiazepines for 4 weeks or less rarely results in significant withdrawal symptoms. If used for 4 months – 5-10% have withdrawals; 2 years – 25-45%; 6-8years – 75% (Law et al. 2004)
- ★ **The role of flumazenil:** In contrast to barbiturates, the benzodiazepines have a larger margin of safety in overdoses. The ratio of lethal dose to effective dose is about 200 to 1 or higher. When benzodiazepines are taken in overdose with alcohol or other antidepressants, this can be more serious. In such cases, death is usually due to respiratory depression. Flumazenil is a specific benzodiazepine antagonist used in A&E/ICU to reverse the effects of the benzodiazepines.
- ★ **Management of dependence in non-abusing patients** with early/mild dependence, minimal interventions such as advisory letters or short courses of relaxation. Where dependence is established, graded discontinuation of prescribed benzodiazepine should be tried; the goal must be cessation after safe withdrawal. Overall 66% may stop in the short term though the long-term outcome is unknown. Additional psychological therapies

do not appear to increase the effectiveness of graded discontinuation but can be considered. Additional pharmacological treatments such as buspirone, antidepressants and anticonvulsants have been repeatedly shown to have no meaningful effect.

- ★ **Management of illicit drug users** is less clear with no robust evidence to support maintenance prescribing. Carbamazepine may be used instead of benzodiazepines to control withdrawal symptoms from high doses of benzodiazepines

GHB: Gamma hydroxy butyrate

- ★ GHB is a naturally occurring substance in the brain. It was synthesized as an anesthetic initially but banned later due to rampant abuse by body builders, as it was thought to boost growth hormone release. It is also abused as a date rape drug (being a colourless liquid, it is easily concealed). It is sold as a powder that is reconstituted or in a concentrated shampoo-like form.
- ★ GHB is also abused for its euphoric effects that are short lived and is used as a substitute for ecstasy in dance parties (raves). It goes by street names of Georgia Home Boy and liquid ecstasy.
- ★ **Pharmacology:** GHB has a GABA-like action; also inhibits dopamine release at a low dose but boosts dopamine availability on chronic use. GHB can also induce the release of noradrenaline in the hypothalamus. It causes respiratory depression and coma, especially if mixed with alcohol. But the person may remain combative despite respiratory depression. Can produce bradycardia in 1/3rd of users.
- ★ A compound derived from GHB called **sodium oxybate** is licensed for named patient use in USA for cataplexy associated with **narcolepsy**.
- ★ In ICD and DSM, GHB abuse is described under the abuse of **sedative-hypnotic** drugs. A **withdrawal syndrome** is reported in heavy users with features very similar to alcohol and benzodiazepine withdrawal. In mild cases, it is limited to insomnia, tremors and anxiety. But may extend to paranoia, hallucinations, psychotic behaviour and extreme agitation in some. These effects start in 12 hours and can last up to 12 days.
- ★ Routine toxicology screening does not detect GHB. Management of intoxication is primarily supportive. Only anecdotal evidence exists in managing withdrawal syndrome – lorazepam, diazepam or haloperidol could be used. Aspiration pneumonia is a common cause of death among GHB users.

Stimulants:

A. Amphetamines:

- ★ 22% of 16-29 year old in UK have used amphetamines at least once. 10% of those who present to addiction services have a primary amphetamine-related problem.
- ★ **Pharmacology:** Amphetamines block catecholamine (DA & NEN especially) reuptake and stimulate their release from vesicles. Typically amphetamines are used to stimulate performance and induce a euphoric feeling by students, truck drivers, etc. Oral, IV and insufflation (snorting) are the possible routes of use. Amphetamine is also called whiz or speed.
- ★ Many synthetic forms are now available. Some amphetamines such as phenylpropanolamine (PPA) can exacerbate hypertension, toxic psychosis, and can cause intestinal infarction and death. Methamphetamine is a potent form that can be inhaled, smoked, or injected intravenously. Ice is a particularly strong purer form – smoked or injected.
- ★ **Intoxication effects** include tachycardia or bradycardia (sometimes these arrhythmias can be fatal), pupillary dilation, elevated or lowered blood pressure, perspiration or chills, nausea or vomiting, evidence of weight loss, psychomotor agitation or retardation, muscular weakness, respiratory depression, chest pain, or cardiac arrhythmias, confusion, seizures, dyskinesias, dystonias, or coma, psychological effects such as euphoria, changes in sociability, anxiety, tension, stereotyped behaviours and impaired judgment.
- ★ **Withdrawal effects** include dysphoric mood (crash) sometimes with suicidal ideation. fatigue, vivid, unpleasant dreams, hypersomnia, increased appetite, psychomotor retardation and small pupils.
- ★ **Tolerance** develops over the due course of use; amounts several 100-fold greater than the amount originally used may eventually be ingested or injected. Tolerance to various effects develops unequally. Tachycardia, euphoria, anorexic effects and increased alertness diminish. It is noted that acute tolerance develops to the subjective but not to the cardiopressor (hypertensive) effects of d-amphetamine in healthy subjects. This finding implies that individuals who repeatedly administer the drug to maintain certain levels of subjective effects may increase plasma drug levels and physiologic effects to toxic levels.

B. MDMA (3, 4-methylene-dioxymethamphetamine):

- ★ MDMA is a substituted amphetamine that produces subjective effects resembling those of amphetamine and LSD (lysergic acid diethylamide). So it is partially hallucinogenic and partially stimulant. It is often used for extended dance parties called raves among adolescents. In the past several psychiatrists used it as an adjunct to psychotherapy as it can increase empathizing ability albeit for a short while.

- ★ **Pharmacology:** It has two optical isomers: the R(-) isomers produce LSD-like effects, and the S(+) isomers produce amphetamine-like effects. The LSD-like action is due to the capacity to release serotonin from terminals – in the long term this damages the serotonin nerves irreversibly and results in depression. Lower doses are more stimulant like; higher doses produce more hallucinogenic or psychotomimetic effects. It can be consumed orally, injected or snorted. Both tachyphylaxis (quick loss of activity after short repeated use) and some tolerance are reported.
- ★ MDMA abstinence-associated **withdrawal symptoms** include fatigue, loss of appetite, depression, anxiety, and trouble concentrating. This happens on the same day or 2 days after the last drug use, the most common mental or mood complaints are difficulty concentrating, depression, anxiety and fatigue. These symptoms strongly resemble a miniature "crash" that is seen as a withdrawal reaction after the use of stimulants.
- ★ Harm reduction advice regarding maintaining hydration and avoiding overheating during use

C. Khat (Cathinone):

- ★ The fresh leaves of *Catha edulis* from East Africa (Somalia, Eritrea), and Yemen are used as a stimulant for their amphetamine-like effects induced by the main ingredient cathinone.
- ★ Cathinone has most of the CNS and peripheral actions of amphetamine. It is buccally absorbed after chewing the leaf and being an alkaloid, gets metabolized rapidly with low chances of toxicity.
- ★ Synthetic methcathinone is also available for illicit use in many countries.

D. Cocaine:

- ★ Cocaine is an alkaloid derived from the shrub *Erythroxylon coca*. It is a powerful positive reinforcer and a strong dependence producer. It has a potent dopamine reuptake blockade effect.
- ★ It can be injected, smoked or snorted. Snorting can cause nasal septal perforation due to local anaesthetic effect with repeated trauma, and vasoconstriction and ischemic necrosis.
- ★ **Pharmacology:** Street cocaine can be freebased (released free from the base e.g. sodium bicarbonate which is mixed with it) to produce a purer form called crack which when inhaled acts as rapidly as IV use. The duration of cocaine's euphoric effects ('high') depends upon the route of administration. The faster the absorption, the more intense the 'high'. Also, the faster the absorption, the shorter the duration of action. The high from snorting is relatively slow in onset, and may last 15 to 30 minutes while that from smoking or mainlining may last 5 to 10 minutes. Inhalation (smoking) or intravenous injections cause a very rapid onset of action.

- ★ At least 60% of cocaine users admit to mental health problems when surveyed. Among these 18% have psychotic experiences.
- ★ **Intoxication:** There is increased energy, increased confidence, heightened self-esteem, and perceived improvement on mental and physical tasks, euphoria, and diminished need for sleep. With high doses agitation, irritability, impaired judgment, impulsive aggression, some symptoms of mania with associated tachycardia, hypertension, and mydriasis are noted. In some, a sensation of bugs crawling beneath the skin (formication) has been reported when under the influence.
- ★ **Withdrawal:** Effects of cocaine are short-lived - rapid cessation occurs due to rapid metabolism resulting in withdrawal effects that can lead to repeated use. A characteristic feature is intense cravings for cocaine with a striking lack of much physical withdrawal symptoms. Dysphoria (crash), anhedonia, anxiety, irritability hypersomnolence, and sometimes agitation are seen upon withdrawal. These effects usually end within 18 hours though in heavy, dependent users this can last up to a week, peaking in 3 days.
- ★ **Physical adverse effects of cocaine includes**
 - Nasal perforation on snorting.
 - Nonhemorrhagic cerebral infarctions.
 - Subarachnoid, intraparenchymal, and intraventricular hemorrhages.
 - TIAs
 - Seizures (3 to 8% of A&E visits)
 - Myocardial infarctions and arrhythmias

E. Caffeine:

- ★ **Pharmacology:**
 - Caffeine is a methylxanthine like theophylline. The half-life of caffeine in the human body is 3 - 10 hours and the time of peak concentration is 30 to 60 minutes.
 - Caffeine readily crosses the blood-brain barrier and acts as an antagonist of the **adenosine** receptors, and so increase the intraneuronal cAMP. At high doses, it can affect dopamine and noradrenergic neurons. Caffeine in nontolerant individuals may reduce **GABAergic** activity in cerebral cortex and striatum, but tolerance to caffeine brings back the GABAergic activity to the control value.
 - A single cup has low to moderate doses of caffeine (i.e., 20 to 200 mg) that increases alertness. 300 to 800 mg produces unpleasant anxiety and nervousness.
 - Caffeine results in global cerebral **vasoconstriction** and a reduction in cerebral blood flow; this may not occur in those over 65yrs. Tolerance does not develop to these vasoconstrictive effects; blood flow shows a rebound increase after withdrawal.

- ★ **Intoxication:** Caffeine intake in excess of 250 mg (>2-3 cups at once) can produce restlessness, nervousness, excitement, insomnia, flushed face, diuresis, gastrointestinal disturbance, muscle twitching, rambling flow of thought and speech, tachycardia or cardiac arrhythmia, periods of inexhaustibility and psychomotor agitation.
- ★ **Withdrawal** is not recognized in DSM. But research criteria indicates that headache, marked fatigue or drowsiness, marked anxiety or depression and nausea or vomiting can be seen even at low doses from 100mg/day, beginning within 12 hours, peaking at 24-48 hours and lasting up to 1 week.

Hallucinogens:

- ★ Hallucinogens are also called psychedelics or psychotomimetics.
- ★ Naturally occurring hallucinogens include **psilocybin** (from mushrooms) and **mescaline**. **Magic mushrooms** especially the **Liberty Cap** (*Psilocybe semilanceata*) is popular in UK. They may be eaten raw or cooked and prepared as a concoction. Small doses cause euphoria, while larger doses (>25 mushrooms) act like LSD. They are not associated with dependence or withdrawal; tolerance develops quite quickly.
- ★ **LSD** is a classic synthetic hallucinogen, synthesized in 1938 by Albert Hoffman. LSD is a synthetic base derived from the ergot alkaloids. It is also noted in rye fungus and was responsible for lethal outbreaks of St. Anthony's fire in the middle Ages. The compounds are also present in morning glory seeds in low concentrations. It is the most commonly used hallucinogen in the UK. LSD has been commonly distributed as blotter acid. Sheets of paper are soaked with LSD, dried and perforated into small squares. Popular designs are stamped on the paper.
- ★ **LSD** acts via sigma opioid and aspartate receptors as well as serotonergic effects. It can cause acute confusion, visual sensory distortions, aggression, and sudden severe violence. Psychosis can also result from its use. Tolerance to the sensory and other psychological effects is evident as soon as the second or third day of successive LSD use. 4 - 6 days free of LSD is necessary to lose significant tolerance. Cross-tolerance among mescaline, psilocybin, and LSD occurs, but not between amphetamine and LSD. Both physical and psychological dependence on hallucinogens are rare and not seen in present day clinical practice. This may be due to lack of reliable euphoria from hallucinogen use.
- ★ **Phencyclidine** (PCP) or **angel dust** is also a synthetic agent not commonly found in the UK except as a contaminant of other drugs. PCP is usually smoked or snorted.
- ★ **Ketamine** is structurally similar to phencyclidine and is used as a veterinary anaesthetic. It does not depress the reticular activating system (thus does not knock out consciousness completely) but reduces the cortical awareness of painful stimuli. It is taken as a sniffed powder termed as 'K'.
- ★ **Hallucinogen intoxication** is characterized by marked anxiety or depression, ideas of reference, fear of losing one's mind, paranoid ideation, perceptual changes e.g., subjective intensification of perceptions, depersonalization, derealization, illusions, hallucinations, synesthesias in a state of full alertness, pupillary dilation, tachycardia, sweating, palpitations, blurring of vision, tremors, and incoordination. The usual sequence of changes follows a pattern of somatic symptoms appearing first, then mood and perceptual changes, and, finally,

psychological changes. In addition, **PCP intoxication** could cause the following additional features: vertical or horizontal *nystagmus*, numbness or diminished *responsiveness to pain*, ataxia and *dysarthria* with *muscle rigidity*. Pupil size is variable with PCP use, depending on the degree of hypoxia and intoxication.

Nicotine:

- ★ Nicotine is one of the most addictive and widely used substances worldwide. Nicotine dependence is recognized in ICD-10 and DSM-IV as a psychiatric disorder. 80% of cigarette smokers are nicotine dependent.
- ★ **Pharmacology:**
 - Nicotine stimulates central nicotinic acetylcholine receptors and improves alertness.
 - Polycyclic hydrocarbons in cigarette smoke are known to stimulate the hepatic microsomal enzymes system, particularly CYP1A2. Smoking can lower the blood levels of some drugs by up to 50%. This can affect both efficacy and side effects. The drugs most likely to be affected are clozapine, olanzapine, chlorpromazine, fluphenazine, haloperidol, tricyclic antidepressants, mirtazapine, fluvoxamine and propranolol.
- ★ **Intoxication:** Nicotine intoxication is not coded separately in classification systems. The most important reason for therapeutic intervention in nicotine users is physical ill-health, especially cardiovascular and respiratory diseases.
- ★ **Withdrawal symptoms:** These are recognized in DSM. The features of withdrawal reaction are dysphoric or depressed mood, insomnia, irritability, frustration, or anger, anxiety, difficulty concentrating, restlessness, decreased heart rate and increased appetite or weight gain. These symptoms begin a few hours after the last cigarette, peak 2 to 3 days after quitting, and become less intense over 1 to 3 weeks. Smokers also report *cravings* for cigarettes, which can last for a longer period of time. The mean weight gain after smoking cessation is approximately 10 lb, and women and heavy smokers are at greatest risk for large weight gains

CLUB DRUGS

National Institute on Drug Abuse has listed some drugs as 'club drugs' according to their popularity in dance clubs and other party venues.

These agents have diverse mechanisms of action, and include MDMA, Ecstasy, Rohypnol, GHB, Ketamine, LSD.

*GHB, flunitrazepam, rohypnol, and ketamine are also called **date rape drugs**. These agents produce sedation with strong amnesia. Thus they are frequent culprits in drugging / 'spiking' for the purpose of theft or sexual assault.*

Inhalants:

- ★ Inhalant substances are easily available, legal, and inexpensive – contributes to nearly 1% of all substance related deaths. The most common profile is that of an adolescent who skips classes at school.
- ★ Often the substances inhaled are solvents for glues and adhesives; propellants for aerosol paint sprays, hair sprays, thinners (e.g., paint products and typing correction fluids); and fuels such as gasoline or petrol.
- ★ **Pharmacology:** Inhalants generally act as CNS depressants; the concentrations of many inhalant substances in the blood are increased when used in combination with alcohol. The effects appear within 5 minutes of inhalation of vapours and can last for 30 minutes to several hours. Tolerance for inhalants has been reported.
- ★ Signs of **intoxication** include dizziness, nystagmus, incoordination, slurred speech, unsteady gait, lethargy, depressed reflexes, psychomotor retardation, tremor, generalized muscle weakness, blurred vision or diplopia, stupor or coma and euphoria. A recent inhalant user may have *rashes around nose and mouth*; unusual breath odours; the residue of the inhalant substances on body or clothes; and *signs of ocular and oropharyngeal irritation*.
- ★ No dependence syndrome has been described; no reliable withdrawal syndrome has been consistently recorded as per DSM.

4. Prescribing controlled drugs

The Misuse of Drugs Regulations 2001 define the authorised persons who can supply and possess controlled drugs in their professional capacities. Accordingly, drugs are divided into five schedules.

2001 MISUSE OF DRUGS REGULATIONS		
Schedule	Examples	Regulations
1	Coca leaf, cannabis, LSD, mescaline	No recognized medicinal use. Supply is limited to research or other special purposes judged to be in the public interest. Requires Home Office licence to possess.
2	Diamorphine, dipipanone, morphine, remifentanyl, pethidine, secobarbital, glutethimide, amphetamine and cocaine	Subject to special prescription requirements and safe custody requirements (with the exception of secobarbital). Stock drugs must be recorded in a register that meets the requirements of the 2001 Regulations, and drug stock must only be destroyed in the presence of an appropriately authorized person.
3	The barbiturates (except secobarbital), buprenorphine, diethylpropion, mazindol, meprobamate, pentazocine, phentermine, and temazepam	These drugs are subject to the special prescription requirements (except for temazepam) but not to the safe custody requirements (except for buprenorphine, diethylpropion, flunitrazepam and temazepam) or to the need to keep a register, although there are requirements for the retention of invoices for 2 years
4	Part 1	These drugs are not subject to the special prescription requirements or to safe custody requirements. There is no need to keep a register, although there are requirements for the retention of invoices for 2 years
	Part 2	
5	Weak preparations of drugs usually in other schedules, for example, morphine, codeine	Exempt from all controlled drug regulations except the need to keep invoices for at least 2 years

All controlled drug prescriptions should have .

- ★ The patient's full name, address and age
- ★ If a patient is homeless, 'no fixed abode' is an acceptable address
- ★ The name and form of the drug MUST be written
- ★ The strength of the preparation, where appropriate
- ★ The dose to be taken MUST be written
- ★ The total quantity of the preparation, or the number of dose units, to be supplied in both words and figures
- ★ A patient identifier number (e.g. NHS number) should be included on prescriptions for controlled drugs
- ★ Prescriptions must be signed by the prescriber with their usual signature (this must be handwritten) along with GMC number as a good practice
- ★ The validity period of prescriptions for Schedule 1, 2, 3 and 4 controlled drugs have been restricted to 28 days.
- ★ Schedule 2 and 3 drugs cannot be prescribed on repeat prescriptions or under repeat dispensing schemes.
- ★ Patients ideally should collect the controlled drug in person after showing their identification on the first occasion and signing the back of the prescription form.
- ★ Substitute opioids should be prescribed in daily instalments whenever required.

Prescriptions of instalments must specify

- The number of instalments
- The interval between instalments,
- Instructions for supplies at weekends or bank holidays
- The total quantity to provide treatment for a period (this must not be exceeding 14 days generally)
- The quantity to be supplied in each instalment along with the duration of the instalments to be set out on the prescription, for example 'dispense daily for fourteen days starting on 3rd September 2015'.

A quick review of addiction pharmacology

DRUG	MECHANISM
Alcohol	Intercalates into the fluid cell membrane; decreases NMDA sensitivity; increases GABA sensitivity; down-regulates calcium channels; up-regulates nicotine receptor gated sodium channels.
Amphetamine	Acts via releasing stored monoamines especially noradrenaline and dopamine. Hence a central sympathomimetic.
Buprenorphine	Partial opioid agonist. Lower doses – mild agonism; higher doses – antagonistic effects.
Cannabis	Acts via cannabinoid receptors. CB1 is central and activated by 11OH tetra hydro cannabinoid. This inhibits GABA tone in the substantia nigra and other areas. May be related to increased dopamine activity at reward centres. CB2 is peripheral immune-related and seen in spleen and thymus. (Endogenous cannabinoids called anandamides are derived from arachidonic acid; their function is unclear)
Clonidine, lofexidine	Presynaptic alpha 2 agonist – reduces central sympathetic tone. Opioid receptors on locus coeruleus projections reduce noradrenergic tone on long-term use. The cellular machinery compensates via up-regulation of adenylate cyclase and maintains sympathetic tone in a chronic user. Sudden withdrawal leads to increased adrenergic firing rate (withdrawal symptoms); hence alpha 2 autoreceptor stimulation which reduces central sympathetic tone helps in opioid withdrawal.
Dexfenfluramine & Fenfluramine	Produce massive serotonin release from nerve endings. [Fen-Phen was an off-label combination of fenfluramine and phentermine used for promoting weight loss but fenfluramine (and dexfenfluramine) was withdrawn due to irreversible serotonergic damage, valvular regurgitation and pulmonary fibrosis].
Disulfiram	Inhibits aldehyde dehydrogenase. Leads to accumulation of acetaldehyde if alcohol is consumed producing unpleasant reactions.
Levomethadyl acetate (LAAM)	Long-acting opioid agonist; potentially similar use as methadone. Withdrawn due to prolonged QT and torsades de pointes. Pure mu agonist.
LSD	5HT _{2A} partial agonism producing hallucinogenic effect
MDMA	Has 2 isomers → R(-) isomers produce LSD-like effects and the S(+) isomers have amphetamine-like properties LSD-like action is mediated via serotonin release from presynaptic neurons. In the long term, this can damage serotonergic tracts irreversibly.
Methadone	Opioid receptor agonist. Longer acting than heroin and orally available. Pure mu agonist.
Naloxone	Short-acting opioid mu antagonist
Naltrexone	Longer acting opioid mu antagonist
Phencyclidine	Noncompetitive NMDA antagonist similar to ketamine; also binds to sigma receptors
Varenicline	Varenicline (Champix) is a partial agonist at the $\alpha 4\beta 2$ unit of nicotinic acetylcholine receptor. It assists smoking cessation by relieving nicotine withdrawal symptoms and reducing the rewarding properties of nicotine.

Notes prepared using excerpts from

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SPMM
Smart
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Addiction Psychiatry II

Paper B

Syllabic content 11.5 to 10

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1. Psychiatric effects of drugs and alcohol

Severely mentally ill (SMI) have high rates of substance use disorders (SUD) but the reasons for this is unclear. Studying the temporal sequence of what starts first is not useful to answer this question because 1. The first drink/drug exposure happens at a constant age in general population irrespective of the later development of mental illness. 2. It is hard to say when a patient really qualifies for a 'substance use disorder' diagnosis in his drug-use career.

- ★ A **common factor model** proposed a common factor such as genetic vulnerability, antisocial personality traits or low social status could predispose to both illnesses. Patients with dual diagnosis have more relatives with a substance use disorder. But relatives of substance-free but severely mentally ill patients (SMI) do not have an increase in substance use disorder.
- ★ The **secondary use model** proposes that patients use substances in the first place to alleviate dysphoria or to self-medicate symptoms and reduce social isolation.
- ★ The **supersensitivity model** proposes that mentally ill patients are unusually sensitive to negative social and health consequences of substance exposure leading to a diagnosis of SUD.
- ★ The **secondary illness model** posits that SUD *leads to* mental illness by a mechanism similar to kindling or behavioural sensitisation. This model is popular with the lay public.

A. Alcohol-related neuropsychiatric problems:

★ **Pathological intoxication:**

- It is also called **idiosyncratic** alcohol intoxication – it is not coded under DSM-V.
- It is a severe behavioural reaction developing rapidly after consumption of a small amount of alcohol. Confusion, hallucinations, high psychomotor agitation, impulsive and aggressive behaviour with risk to self or others is noted; this may last for few hours and terminate in a prolonged sleep.
- It is unclear what factors predispose to such reaction – high levels of underlying anxiety, brain damage, advancing age, using sedative-hypnotic drugs, and feeling fatigued are proposed. It is sometimes used as a forensic defence.
- Physical restraint and IM haloperidol may be required.

★ **Alcoholic hallucinosis:**

- The DSM has a diagnostic criterion for **alcohol-induced psychotic disorders** (popularly called alcoholic hallucinosis) and further allows the specification of onset (during intoxication or withdrawal).
- The most common hallucinations are unstructured sounds or voices that may be characteristically malign and threatening. The hallucinations usually last less than a week,

when patients believe in the hallucinations though afterwards they may realise the untrue nature. These usually appear in persons abusing alcohol for a long time.

- Delusions are usually secondary interpretations of the hallucinations. The patient is usually distressed, anxious, and restless. But unlike delirium tremens, those with alcoholic hallucinosis have a clear sensorium.
- As an empirical timescale, hallucinations lasting for more than 6 months should initiate suspicion of other psychoses such as schizophrenia.
- It is believed that 'release' of symptoms can occur due to alcohol use in someone with a vulnerability to develop schizophrenia.
- Differences from schizophrenia include:
 - Atypical or late age of onset of psychotic symptoms
 - Onset of alcohol drinking clearly preceding the onset
 - Remission of psychotic episodes during abstinence.
 - Lack of thought disorder and affect incongruence.
- The hallucinations respond well and rapidly to antipsychotics. The prognosis is good provided that the person remains abstinent. But the evidence base for treating alcoholic hallucinosis is sparse.

★ Alcoholic blackouts

- Alcohol-related blackouts are not included in DSM / ICD10.
- Blackouts are discrete episodes of anterograde amnesia that occur in association with alcohol intoxication.
- Alcohol blocks the consolidation of new memories into old memories at the hippocampus. During a blackout (at the time of intoxication), remote memory is intact but patients experience specific short-term episodic memory deficit (they can talk about their childhood, but cannot remember what topic the conversation is about or how did they come to where they were). They can even perform complicated tasks but later will not remember these. The memory gap usually lasts for hours, rarely a day or more.
- **Alcoholic dementia:** The established types of dementia associated with alcoholism are WE/KS, Marchiafava-Bignami, hepato- cerebral degeneration and vitamin deficiency dementias. In addition, some favour the existence of persistent dementia attributable to the direct toxic effects of alcohol on the brain called **primary alcoholic dementia**. But clear delineation of such an entity is elusive as no distinctive, well-defined pathological basis is established for this diagnosis. Nevertheless, the atrophy and other alcohol-related brain changes have been shown to be **partially reversible** with abstinence in neuroimaging studies.

★ **Wernicke's encephalopathy (WE):** This is an acute condition characterized by the well-known triad of ophthalmoplegia, ataxia global confusional state.

- Classic triad of confusion, ataxia and nystagmus is only present in 10% of patients and 80% of patients are not diagnosed before post mortem. The eye signs are usually bilateral but not necessarily symmetrical A high index of suspicion is needed as it rarely presents with classical signs and symptoms
- WE is directly caused by thiamine deficiency, due to inadequate dietary intake, reduced gastrointestinal absorption, decreased hepatic storage, and impaired utilization.
- Not every thiamine-deficient alcoholic develops WE. A vulnerability related to abnormal thiamine-dependent transketolase, may result in reduced affinity for sparsely available thiamine. Patients with WE show gliosis and small haemorrhages in periventricular and periaqueductal structures especially, the mammillary bodies, hypothalamus, mediodorsal thalamic nucleus, colliculi, and midbrain tegmentum. Neuropathological lesions are often symmetrical
- In some case series, nearly one-eighths, i.e. 12.5% of all chronic alcoholics exhibited signs of Wernicke's encephalopathy.
- MRI is currently the most valuable diagnostic tool to confirm a diagnosis of Wernicke's encephalopathy. MRI has a sensitivity of only 53%, but its high specificity of 93% means that it can be used to confirm a suspected diagnosis (Sechi & Serra 2007, The Lancet Neurology: 6, 5, 442-455)
- No controlled trials exist for treatment recommendations.
 - WE is initially reversible but, if untreated or with inadequate thiamine replenishment can result in irreversible Korsakoff's syndrome in 84% of survivors. Untreated WE is associated with approximately 20% mortality
 - Thiamine should be given before any i.v. glucose. As glucose infusion exacerbates thiamine deficiency biochemically.
 - In healthy uncomplicated alcohol dependent/heavy drinkers (i.e. at low risk), oral thiamine should be given at a minimum dose of 300 mg/day during detoxification
 - If the patient is at high risk of WE, prophylactic treatment should be given, using 250 mg thiamine (one pair of ampoules Pabrinex) i.m. or i.v. once daily for 3–5 days. Pabrinex also contains nicotinamide, pyridoxine (vitamin B6), riboflavin (vitamin B2) and vitamin C.
 - If WE is suspected or established, parenteral thiamine (i.m. or i.v.) of > 500 mg should be given for 3–5 days (e.g. two pairs of ampoules Pabrinex three times a day for 3 days, followed by one pair of ampoules once daily for 3–5 days) depending on response.
 - Ophthalmoplegia responds within hours, but cognitive impairment takes longer.

★ **Korsakoff's syndrome (KS)**

- About 80 per cent of alcoholic patients recovering from Wernicke's encephalopathy later develop Korsakoff's amnesic syndrome.
- It is characterized by marked deficits in anterograde and to some extent retrograde episodic memory, apathy, an intact sensorium, and relative preservation of other intellectual abilities such as attention, procedural memory and working memory.
- The physiological processes that underlie the storage of information in a more permanent form (consolidation by 'transfer' of information from primary to secondary memory) are thought to be affected. Consistent with this hypothesis, digit span test scores and other measures of working memory are characteristically intact in patients with Korsakoff's syndrome.
- Korsakoff's amnesic syndrome can occur without antecedent WE. Further, repeated subclinical (undiagnosed) episodes of WEs might culminate in Korsakoff's amnesic syndrome.
- Diagnosis of KS correlates best with the presence of lesions in the dorsomedial thalamus. But lesions in the mammillary bodies, the mammillothalamic tract and the anterior thalamus may be more important to memory dysfunction than lesions in the medial dorsal nucleus of the thalamus.
- Confabulation occurs due to a double lesion – frontal and diencephalic deficits.

★ Cerebellar degeneration

- Degeneration of Purkinje cells in the cerebellar cortex (by 10–35%) due to alcohol-induced damage results in cerebellar atrophy.
- Has been reported to occur in about 40% of chronic alcoholics
- Limb ataxia and dysarthria occur more often than in Wernicke's encephalopathy, whereas nystagmus is rare.
- Considerable numbers of alcoholic patients with MRI cerebellar atrophy are not ataxic on examination. Cerebellar ataxia in alcoholics does not correlate directly with the daily, annual, or lifetime consumption of ethanol.
- Impaired cerebellar function improves significantly when abstinence is maintained.

★ Hepato- cerebral degeneration

- Hepatic **encephalopathy** due to liver failure is characterized by altered sensorium, frontal release signs, 'metabolic' *flapping tremor*, hyperreflexia, and extensor plantar responses.
- High mortality rates are noted; in those who make a partial recovery, a discrete neurological deficit characterised by tremor, choreoathetosis, dysarthria, gait ataxia, and dementia results. This is called as **hepatocerebral degeneration**.

- Enlargement and *proliferation of protoplasmic astrocytes* in the basal ganglia, thalamus, red nucleus, pons, and cerebellum, in the absence of neuronal loss or other glial changes, is noted.

★ **Marchiafava–Bignami syndrome**

- Carducci first described it in 1898 in Italian red wine drinkers and later by Marchiafava and Bignami expanded the description in 1903. It occurs predominantly in malnourished alcoholics and is reported more often in male than female drinkers.
- About 250 cases have been reported in the medical literature, but it is likely that its incidence is higher since this diagnosis might easily be missed.
- Demyelination of the corpus callosum and adjacent subcortical white matter (Extra- pontine myelinolysis).
- Dementia, spasticity, dysarthria, gait changes occur.
- Patients may lapse into coma and die, survive for many years with a dementia
- Reported to be unusually high in Italian red wine drinkers.
- 2 clinical types are identified (Heinrich, 2004): Type A had predominant features of coma and stupor. This subtype is associated with a high prevalence of pyramidal tract symptoms. Radiologic features include involvement of the entire corpus callosum. Type B is characterized by normal or mildly impaired mental status. Radiologic features are partial or focal callosal lesions.

★ **Central pontine myelinolysis (CPM)**

- Adams et al. first described CPM in 1959.
- It is characterized by demyelination that involves the central portion of the base of the pons.
- Clinical manifestations include of pain sensation in the limbs, bulbar palsy, quadriplegia, disordered eye movements, vomiting, confusion, and coma or locked-in syndrome.
- It also occurs in non-alcoholics with liver disease including Wilson's disease, malnutrition, anorexia, burns, cancer, Addison's disease, and severe electrolyte disorders such as thiazide-induced hyponatraemia.
- Myelinolytic lesions can be induced experimentally by rapid correction of chronic hyponatraemia.

★ **Pancreatitis:**

- Alcoholic pancreatitis is associated with hypocalcemia. When the pancreas is damaged, free fatty acids are generated by the action of pancreatic lipase. There are insoluble calcium salts

present in the pancreas and the free fatty acids avidly chelate the salts resulting in calcium deposition in the retroperitoneum with associated back pain seen in pancreatitis.

- In addition, hypoalbuminemia may be part of the clinical picture so that there is a reduction in total serum calcium. If there is concomitant emesis or poor nutritional status, hypomagnesemia may augment the problem.
- In pancreatitis-induced hypocalcemia, parathyroid hormone levels can be normal, suppressed or elevated (in severe hypocalcemia).
- In the treatment of these patients, parenteral calcium and magnesium replacements are indicated. Vitamin D status should be assessed to rule out malabsorption or nutritional deficiencies.
- Chronic pancreatitis may also produce hyperglycaemic status and polyuria/nocturia and is associated with a high incidence of depression.

B. Neuropsychiatric problems due to illicit drugs:

- ★ **Stimulants-related psychosis:** The hallmark of amphetamine-induced psychosis is the paranoia. *Sensitisation* is suggested to be the biological mechanism behind stimulant-induced psychosis. Clinical features are very similar to schizophrenia but for the following features in stimulant-induced psychosis:
 - The absence of prominent negative symptoms.
 - The predominance of visual hallucinations.
 - Generally appropriate affect.
 - Associated with hyperactivity.
 - Disinhibited sexual behaviour.
 - Confusion and incoherence.
 - Almost no formal thought disturbance.

Often the rapid resolution of the symptoms in a few days or a positive finding in a urine drug screen test is the only means of making a correct diagnosis. The treatment of choice is the short-term use of an antipsychotic medication such as **haloperidol**.

- ★ **Cannabis-related syndromes:** Florid psychosis referred to as **hemp insanity** occurs after high doses of high-potency cannabis use. Cannabis use rarely causes a **bad-trip** similar to hallucinogen intoxication. The long-term use of cannabis may produce subtle forms of **cognitive impairment** in memory, attention and in the integration of complex information. A controversial **amotivational syndrome** has been associated with long-term heavy use. Patients show loss of motivation to persist in a task that requires prolonged attention or effort. They may appear apathetic and anergic and may be gaining weight. Mild attentional problems and impulsivity are common in children born to mothers actively consuming marijuana through **pregnancy**.

★ **Hallucinogen-related syndromes:**

- **'Bad trip'** is the most common adverse effect of LSD and related substances. It resembles an acute panic reaction but sometimes it could be more severe, lasting for a longer period than a panic attack. A bad trip can occasionally produce true psychotic symptoms. It generally ends when the immediate effects of the hallucinogen wear off, but its course is variable.
- Long after ingesting a hallucinogen, a person can experience a flashback of hallucinogenic symptoms. This syndrome is diagnosed as **hallucinogen persisting perception disorder** in DSM-V. This reexperiencing of perceptual symptoms following cessation of hallucinogen use is characterised by seeing geometric hallucinations, false perceptions of movement in the peripheral visual fields, flashes or intensified colours, trails of images of moving objects, positive afterimages, halos around objects, macropsia, and micropsia. This is also called **flashback phenomenon**. Emotional stress, sensory deprivation or the use of another psychoactive substance, such as alcohol or marijuana can trigger a flashback.

★ **Inhalant-related syndromes:** Chronic inhalant-users may develop diffuse cerebral, cerebellar, and brainstem atrophy with white matter disease, a leukoencephalopathy seen in CT/MRI. The clinical effects may include hearing loss, peripheral neuropathy, cerebellar signs, persisting motor impairment, parkinsonism, apathy, poor concentration, memory loss, visual-spatial dysfunction, impaired processing of linguistic material, and lead encephalopathy in some cases.

2. Prevention & Public Health

A. Risk factors for alcoholism:

- ★ **Social learning model** and **disease model** are popular among addiction specialists. The **social learning model** views alcoholism as a maladaptive behaviour. The disease model views the issue as somewhat *irreversible* but *reduces self-blame*; **loss of control** is a primary concept within the disease model.
- ★ **Sociocultural factors:** A high rate of domestic violence is seen in association with alcohol use. The family systems model considers that the behaviour of dysfunctional family members around drinking, over time, stabilizes the family system forcing families to organise their structure and function around alcohol to preserve a degree of familial 'homeostasis.'
- ★ **Genetic factors:** (see table below). Alcoholism is a common condition and 40% of the general population has a family history of alcoholism. Having multiple family members affected by alcoholism is seen in approximately 7-9% of the population. Several findings support that individual variations in alcohol sensitivity and alcohol metabolism are likely to be important determinants of the risk of developing alcoholism.
- ★ MZ have greater *concordance* than dizygotic; *adoptees* have the risk of alcoholism proportional to the biological family (1.6 to 3.6 times greater in adopted away children of alcoholics). Close family members have 4-fold increased risk of alcoholism.
- ★ *The Virginia Adult Twin Study of Psychiatric and Substance Use Disorders* estimated the heritability of alcohol use disorders to be 0.51 to 0.66 when all definitions of alcoholism were included. Some studies claim that males have the higher familial tendency (50 – 60%) than females (25-30%).
- ★ Dawson et al. in 1992 calculated that the odds of alcohol dependence were increased by 167% (roughly 4 times) in individuals with both a first and second degree relative with alcoholism, 86% (roughly 2 times) in those with an affected first-degree relative, and 45% (roughly 1.5 times) among those with a second or third degree affected relative.

ALCOHOL QUANTITY IN UNITS

A simple method is to multiply ABV in % (alcohol by volume) for a drink by the liters consumed.

Thus two pints (2 X 568ml = 1.136 liters) of beer at 4% ABV contain

$$1.136 \times 4 = 4.5U$$

Safe units

In males – up to 21 U a week; 21 to 49 – hazardous; more than 49 – harmful.

In females – up to 14 U a week; 14-35 hazardous; more than 35 harmful.

- ★ Putative biomarkers: Low 5HT, 5HIAA and MAO are found in impulsive problem drinkers. P300 amplitude is reduced in alcoholics and their children too. Men with a family history of alcoholism show less physiological response (cortisol and sway test) when alcohol is ingested.

Genetic loci	Role	Findings
Chromosomes 4p13-12 (GABRB1) Chromosome 5q33-34 (GABA _A α1, α6, γ2)	Gamma-aminobutyric acid receptor subunits	Linkage regions in association studies
Chromosome 11q23.1	dopamine D2 receptor gene (<i>DRD2</i>) <i>Taq1</i> polymorphism	Controversial; most authors do not believe in significant association
Chromosome 12q24.2	Aldehyde dehydrogenase variants	Single base pair change in exon 12 of the <i>ALDH2</i> gene produces Antabuse like reaction due to reduced ALDH activity. Homozygous in 12% people of South East Asia/Chinese origin conferring protection.
Chromosome 4q22 cluster	Alcohol dehydrogenase polymorphisms	High-activity variants <i>ADH2</i> *2 and <i>ADH3</i> *1 are significantly decreased in people with alcohol dependence at least in the oriental population. They produce less acetaldehyde at a slower pace.

B. Risk factors for substance use:

Sociocultural factors: Several social factors shape the population prevalence of substance use behaviour. Contextual factors such as neighborhood deprivation appear to be strong determinants of cigarette and alcohol use. Family and social network norms and social support are also important in the cessation of drug use. Factors that are consistently identified in association with substance use and alcohol are listed below (Galea et al., 2004)

Smoking	Alcohol use	Illicit substances
• Low school achievement • Young among peer cohort • Poorer relationships with their family • Low household income	• Disruption of family structure • Social networks that use alcohol • Recent immigration • Small-area deprivation	• Peer drug use • Single parenting • Homelessness • Poor educational attainment • Neighborhood disadvantage • Unemployment

C. Prevention strategies:

The evidence base for the effectiveness of prevention strategies is scarce. Nevertheless, public health bodies have advocated approaches that focus on 5 broad domains of risk factors.

Risk Factors	Domain	Preventive measures
Aggressive Behaviour	Individual	Early identification and targeted psychological interventions
Poor parenting, dysfunctional attitudes towards substances	Family	Parental Training & Family Education
Substance exposure	Peer groups	Academic competence, legal policies
Drug availability	School	Legal policies, school-based educational interventions
Poverty and social deprivation	Wider society	Promoting neighbourhood cohesion

D. Screening tools:

- ★ Alcohol use disorders identification test (AUDIT) was designed to be used as a brief structured interview or self-report questionnaire. It is suitable for general practice use. The AUDIT was found to have a sensitivity of 83% among males and 65% among females in primary care setting. Shorter versions are available e.g. The Fast alcohol screening test (FAST) was developed from AUDIT as a shorter version to use in busy A&E and hospital environments.
- ★ CAGE is the most widely used screening tool for alcohol use. But CAGE does not ask about the frequency of alcohol use, levels of consumption or episodes of heavy drinking, all factors that can identify patients in the earlier stages of problem drinking (Nilssen & Cone, 1994). Aertgeerts et al. (2001) conducted a study to select best screening tool in general practice and found CAGE to be insufficient among primary care patients with only 62% sensitivity for males and 54% for females.
- ★ Michigan Alcohol Screening Test MAST was developed for detecting dependent drinkers. It has 25 questions that relate to respondents self-appraisal of problems frequently associated with excessive drinking. The sensitivity of MAST is 86-98% and specificity 81-95%. The major

drawback of MAST is that the focus is on lifetime rather than the current occurrence of alcohol- related problems. Hence, one may miss problem drinking at early stages (Nilssen & Cone, 1994)

- ★ Note that conventional laboratory tests are of no use for detecting alcohol abuse or dependence in a primary care setting. For detecting hazardous and dependent drinking in primary care, the AUDIT seems to be one of the best tools to use.

E. Urine drugs analysis

Substance	Time present in urine
Alcohol	Up to 12 hrs
Amphetamine	Up to 48 hours
Benzodiazepine	3 days (depending on t _{1/2})
Cannabis	Occasional use – up to 3 days. High daily use for long time – up to 4 weeks.
Cocaine	6 – 8 hrs; metabolites up to 2 - 4 days
Codeine	48 hours
Heroin	1 to 3 days
Methadone	3 days or more
Morphine	2 to 3 days
Phencyclidine (PCP)	3 to 8 days
LSD	Less than 24 hours (average users)

Substance	False positive test producer
Phencyclidine	Dextromethorphan
Marijuana metabolites	Ibuprofen
Opiates	Tonic water
Amphetamines	Phenylephrine decongestants

Adapted from Oxford Handbook of Psychiatry & Yinon, J (ed) Advances in Forensic applications of mass spectrometry p. 88

3. Evidence-Based Treatments for Alcohol Use:

A. Alcohol detoxification:

- ★ Many alcohol withdrawal episodes take place without any medical or pharmacological treatment. Successive episodes of alcohol withdrawal are associated with increasing severity and complications such as cognitive impairment. For an alcohol detoxification programme, the intended goals can be
 1. Symptom suppression
 2. Avoiding complications
 3. Completion of regimen with subsequent abstinence
- ★ Community-based detoxification involves daily contact with a nurse to assess withdrawal and monitor for complications. Indications for inpatient detoxification include
 1. Past history or current symptoms of DT or seizure
 2. Psychiatric morbidity with risk of suicide or debilitating physical health
 3. WE/Korsakoff syndrome
 4. Homelessness or social difficulties where detoxification may not be completed successfully.
- ★ Benzodiazepines are better than placebo for alcohol withdrawal for various outcomes including
 - reducing the severity of withdrawal
 - reduction in the incidence of complications (delirium and seizures)
 - completion of regimen and subsequent rehabilitation
- ★ The choice of which benzodiazepine to use depends on the circumstances of treatment; almost all benzodiazepines are shown to be efficacious in alcohol detoxification. Benzodiazepines with a longer half-life are more effective in reducing the incidence of delirium. Longer dose tapering is also required in those with a history of DT or having current DT.
- ★ Substances that are eliminated without phase 1 metabolism such as lorazepam or oxazepam may be suitable when the patient has liver failure. Longer-acting benzodiazepines may be more effective in preventing seizures and delirium, but they can accumulate in the body of the elderly.
- ★ A typical fixed regimen for covering uncomplicated withdrawal is 20 mg q.d.s. of chlordiazepoxide, reducing over approximately 7 days. Additional titration using PRN medication can be used in the first 2 days.
- ★ **'Front-loading' technique** - a loading dose of diazepam is given followed by and further doses every 90 min until light sedation is achieved. No further medication is given, and the long half life of diazepam covers the withdrawal.

- ★ **A symptom-triggered regimen** uses chlordiazepoxide whenever a clinical withdrawal rating scale such as (CIWA– Clinical Institute Withdrawal Assessment Scale; 10 items score 10-20 moderate, >20 –severe, or SAWS – short alcohol withdrawal scale – self-completed; >12 needs treatment) shows significant symptom scores.
 - **Advantages:** Faster control and fewer benzodiazepine doses needed in total. Waiting for withdrawal symptoms to be manifest is safer with regard to avoiding over-sedation and increased confusion in intoxicated users seeking detoxification.
 - **Problems:** As prevention of withdrawal symptoms is the goal this approach may be meaningless in various settings. The distress caused to the patient may reduce his motivation to be abstinent.
- ★ **Chlormethiazole** can also be used, but better avoided due to a higher risk of respiratory depression than other benzodiazepines especially when alcohol is consumed during detoxification. There are also concerns regarding its variable bioavailability, higher addictive potential and ‘street value’.
- ★ **Carbamazepine** might be a first-line alternative to benzodiazepines. Carbamazepine is effective for the whole range of alcohol withdrawal symptoms, including severe withdrawal and is not contra-indicated in liver failure. But it has limited evidence with respect to preventing DTs in alcohol withdrawal.
- ★ **Anticonvulsants** alone are also as effective as benzodiazepines for reducing the risk of or treating seizures. Both drugs when taken together conferred no advantage in seizure prevention. With regard to preventing a secondary seizure during the same withdrawal episode, phenytoin has been shown to be ineffective. However, lorazepam has been shown to be effective.
- ★ **Haloperidol** has been shown in a meta-analysis to reduce signs and symptoms of alcohol withdrawal, but they do not reduce the risk of seizures or delirium, as effectively as benzodiazepines. In fact, they may increase the likelihood of seizures compared with placebo.

B. Relapse prevention:

- ★ Relapse prevention is not always required. 1 in 10 may have spontaneous abstinence due to social changes.
- ★ In rehab / relapse prevention, the goal may be either controlled drinking or abstinence. Factors that may favour one or the other are listed below;

Controlled drinking	Total abstinence
Age less than 40	>40
Early detection of alcohol problems	Long standing
Minimal dependence	Severe/moderate dependence
No medical or psychiatric complications	Complications present

No impulsivity	Impulsive personality
Stable social support	Poor social support
Clear preference for controlled drinking	Ambivalent or chooses abstinence
Partner or family can monitor	Monitoring not possible

★ **Acamprosate:**

- Acamprosate is a taurine derivative. It inhibits glutamatergic NMDA receptor function and balances GABA-glutamate imbalance seen in alcohol dependence. It is hypothesized to suppress the 'urge to drink' in response to learned cues
- Compared to placebo the odds ratio for abstinence at various time intervals (3, 6, 12 months) is around 1.73; a 'number needed to treat' of 11 to prevent one relapse has been calculated. Reduces craving with an NNT of 8.
- Gastrointestinal disturbance (e.g. nausea, diarrhoea) are the most common side effects reported.
- In almost all studies investigating acamprosate some form of psychosocial intervention was also present - acamprosate alone has not been shown to be effective. Currently, it is recommended to start acamprosate as soon after detoxification as possible. Late onset of treatment may result in loss of efficacy. The benefits of acamprosate may continue 1- 2 years even after stopping the drug.
- The addition of disulfiram to acamprosate also improves outcome compared with acamprosate alone.

★ **Naltrexone:**

- Naltrexone (*Adepend*) is an opioid antagonist licensed in the UK (from 2012) for relapse prevention in alcohol dependence. Naltrexone may reduce the pleasurable effects of alcohol by blocking the effects of opioids released by alcohol; this results in reduced stimulation of mesolimbic dopamine reward system.
- Naltrexone has been shown to be superior to placebo in maintaining abstinence, relapse rates, time to first drink, and reduction in number of drinking days, reduction in craving and improvement in GGT (liver function). Naltrexone is more effective than placebo as an adjunct to psychosocial interventions with OR = 1.46 and NNT of 9 to 11.
- Note that in the original naltrexone trials, the main outcome measure was 'time to relapse to heavy drinking' (i.e. > 5 drinks/day or drinking on more than 5 days/week) rather than abstinence. By contrast, the acamprosate trials mostly used complete abstinence (i.e. having no alcohol in 1 day) as their outcome measure. The naltrexone studies, therefore, provide evidence to treat individuals who resume at least some drinking, rather than those who are fully abstinent (Volpicelli et al., 1997). Therefore, the main effect of naltrexone

may be to reduce drinking, or prevent a 'lapse' becoming a full-blown relapse. Naltrexone should not be stopped if drinking resumes.

- Comparing naltrexone and acamprosate head to head, there were no differences between the drugs on a number of different outcomes (% abstinent, % retention) and both were better than placebo.

★ *Nalmefene:*

- **Nalmefene** (also called Selincro) is licensed for reduction of alcohol consumption. Rather uniquely, it can be prescribed on the 'as-needed' basis, with no need for liver monitoring or detoxification/physical withdrawal before use, and is deemed suitable for initiation in community settings.
- It is an opioid receptor modulator, with antagonist activity at μ - and δ -opioid receptors and partial agonist activity at κ -opioid receptors.
- Nalmefene is recommended as an option for reducing alcohol consumption in dependent individuals with "a high drinking risk level (defined as alcohol consumption of more than 60 g per day for men and more than 40 g per day for women, according to the World Health Organization's drinking risk levels) without physical withdrawal symptoms **and** who do not require immediate detoxification". "It should only be prescribed in conjunction with continuous psychosocial support focused on treatment adherence and reducing alcohol consumption **and** be initiated only in patients who continue to have a high drinking risk level 2 weeks after the initial assessment".

★ *Disulfiram:*

- Disulfiram blocks aldehyde dehydrogenase, resulting in the accumulation of acetaldehyde leading to signs and symptoms such as flushing, nausea, vomiting, headache, tachycardia and palpitations if alcohol is consumed.
- This reaction is dose-dependent, and if a large amount of alcohol is consumed, a more severe reaction may ensue leading to hypertension, collapse and even death. This acts as a deterrent (not directly as aversion – in 'aversion', chemicals such as valeric acid or apomorphine are used to create nausea which is paired with alcohol).
- Witnessing, or supervising, the taking of disulfiram is important for its efficacy. Disulfiram reduces the number of drinking days and reduces the quantity of alcohol consumed, but not increase abstinence. There is little evidence to guide how long patients should receive disulfiram but, generally, 3–6 months is advocated, or for as long any benefits are maintained.

- Impulsivity, psychosis, suicidality, severe hepatic dysfunction, peripheral neuropathies, cardiac diseases are contraindications for disulfiram.

★ **SSRIs:**

- SSRIs may improve drinking outcomes in type 1 alcoholism (later age of onset, anxious traits) but may worsen outcome in type 2 alcoholism (early age of onset, family history positive, impulsive/antisocial personality traits).

C. Psychosocial interventions for alcohol dependence:

Project MATCH (Matching alcoholism treatments to client heterogeneity) was a large multisite (9 centres) USA based RCT with a total of 1726 clients. It tested the hypothesis that matching patient characteristics to specific treatments would improve the overall outcome of alcohol dependence. The psychosocial treatments explored were Cognitive Behavioural Coping Skills Therapy (CBT), Twelve-Step Facilitation Therapy (TSF), and Motivational Enhancement Therapy (MET). The project was divided into two parallel, but independent, clinical trials — an out-patient arm and an aftercare arm.

There was little evidence for matching improving overall outcome. AA attendance predicted a better long-term outcome, particularly in those lacking a non-drinking support network. At 1 year, clients low in psychiatric severity (comorbid mental health issues) had more abstinent days after TSF than after CBT. Neither treatment was clearly superior for clients with higher levels of psychiatric severity. At 3 years, clients whose social networks were more supportive of drinking derived greater benefit from TSF treatment than from MET; those with low support for drinking derived more benefit from MET than TSF. This may be because those who drink even without encouragement from drinking friends may need motivational enhancement more than those who drink due to social cues. Readiness-to-change and self-efficacy emerged as the strongest predictors of long-term drinking outcome.

UKATT (United Kingdom Alcohol Treatment trial) was a multicentre (7 centres Birmingham, Cardiff, Leeds), pragmatic effectiveness RCT involving 742 clients with alcohol problems (83% FU rate at 1 year). It compared Motivational Enhancement Therapy and Social Behaviour and Network Therapy in treating alcohol problems. SBNT is carried out over eight sessions each lasting 50 min. Components were drawn from network therapy, behavioural marital therapy, unilateral family therapy, social aspects of the community reinforcement approach, relapse prevention and social skills training. The novel social behaviour and network therapy for alcohol problems did not differ significantly in effectiveness from the proved motivational enhancement therapy

Slattery et al., (2003) reviewed evidence for a number of interventions and models, including CBT, behavioural self-control training (BSCT), motivational enhancement therapy (MET), coping skills training, marital/family therapy, and intensive case management. Their meta-analysis showed that BSCT, MET, marital/family therapy and coping/skills training had similar, though moderate to small beneficial effect sizes.

- ★ **Behavioural self-control training (BSCT):** The major components include placing limits on number of drinks, self-monitoring using a drink diary, using non-alcoholic spacers, using food to fill before / after drink, altering rate of drinking, developing assertiveness to refuse offer of drink and to reward oneself for reaching goals. The user is encouraged to develop awareness of antecedents and to learn alternate coping skills.
- ★ **FRAMES approach:**
 - Feedback of risks
 - Responsibility highlighted
 - Advised to abstain or cut down
 - Menu of alternative options and activities offered
 - Empathic interviewing
 - Self-efficacy is enhanced regularly throughout the intervention.
- ★ **Twelve step facilitation programme (TSF):** A good example is AA. The AA model emphasizes that once somebody becomes an alcoholic, he / she remains to be so. This principle apparently helps by being continuously vigilant and modifies lifestyle accordingly. Current expert consensus recommends that patients should be introduced to AA, and encouraged to attend, but not mandated to attend as compulsion decreases the efficacy of AA. The 12 steps are
 1. Accept powerlessness in front of alcoholism.
 2. Admit only a Greater power can help
 3. Make a decision to turn our to the care of God as you understand Him
 4. Make a searching and fearless moral inventory.
 5. Admit wrongs done to others.
 6. Become entirely ready for removal of these defects.
 7. Ask Him to help now.
 8. Be willing to make amends to all
 9. Make direct amends where possible
 10. Continue personal inventory.
 11. Prayer and meditation.
 12. Practice and preach.
- ★ **Motivational interviewing** is a directive, patient-centred counselling style based on the transtheoretical model of change that aims to help patients explore and resolve their ambivalence about behaviour change. Miller introduced motivational interviewing. The aim of this approach is to create dissonance in drinker until they become willing to consider alternatives to continued alcohol/drug use. The four principles of motivational interviewing
 1. Express empathy (reflective listening)
 2. Develop attitude-behaviour discrepancy

3. Sidestep resistance / rolling with resistance without confrontation
4. Support self-efficacy for change

The **transtheoretical model** (Prochaska & DeClemente) of change breaks down the concept of readiness to change into stages; from not even thinking about it to maintaining change once it is made. This theory complements motivational interview technique. The stages of change in the transtheoretical model

1. *Precontemplation*, not even ready to think about change seriously
2. *Contemplation*, ready to think about change
3. *Determination/preparation*: preparing to make plans for change
4. *Action*, implementing change
5. *Maintenance*, ensuring that the change in behaviour becomes habitual

★ **Relapse prevention –CBT approach:**

- The main aim is to enhance capacity to maintain abstinence.
- This can reduce relapse and if relapse occurs the intensity of relapse may be less severe.
- Note that overinvestment of absolute abstinence itself contributes to the catastrophic nature of relapses – this is called *abstinence violation effect*.
- *The matrix model* uses a combination of CBT, family education, social support, individual counseling, regular testing of alcohol and optional 12 step program.

★ **Residential rehabilitations:**

Various models have been used – the **Minnesota rehab model** uses detoxification, psychoeducation and AA attendance with 4 weeks stay in the ward. This is followed by after care at rehabilitation residences.

Phoenix house is an example of **concept houses**. It works on a therapeutic community model. A slightly confrontational style is adapted, firm feedback is given, and responsibility is emphasized. Residents have a major role in running the house; they get an increase in privileges and responsibilities depending on their progress.

4. Treatments for Opioid Use

- ★ **Treating opioid overdose:** The most important step is to ensure an adequate airway. Naloxone can be given IV at a slow rate 0.8 mg per 70 kg of body weight. The gain in respiratory rate and pupillary dilation usually occurs soon after naloxone administration. But naloxone being a short-acting drug when compared to heroin and methadone, repeated administration may be needed even after symptomatic improvement; else the toxicity can recur even without additional drug intake. In opioid-dependent patients, naloxone may produce signs of withdrawal.
- ★ **Harm reduction advice:** Not to use opiates while alone; Not to use in combination with other drugs; Avoid IV route; Always inject in the direction of blood flow (in veins, this is towards heart); Rotate injection sites; Avoid going for neck, groin etc.; Ensure complete dissolution before injection – else emboli can occur; Use new sterile needles and syringes on each occasion; Use sterile water – avoid lemon juice, which can cause Candida endophthalmitis; Never share needles or equipment with others
- ★ **Treating opioid withdrawal (BAP Guidelines):**

There is a substantial evidence base for three main types of pharmacotherapy:

1. Methadone 2. Buprenorphine and 3. $\alpha 2$ adrenergic agonists (e.g. clonidine and lofexidine).

All are effective in reducing withdrawal symptoms.

The choice of agent may be guided by the following:

1. Patient's choice

2. Desired duration of treatment

- If a short duration of treatment is desirable, $\alpha 2$ adrenergic agonists are preferable to methadone. As lofexidine do not control symptoms such as nausea, additional symptom support using metoclopramide, antidiarrhoeals, etc. may be required. Clonidine is not so popular now due to high rates of hypotension associated with it.
- Buprenorphine can be used for short-term opioid withdrawal and has a better outcome than clonidine; though it is given once daily it is also effective in thrice-weekly dosing because of its slow dissociation from opioid receptors. As it is a partial agonist, abrupt discontinuation does not produce many withdrawal symptoms, unlike methadone or heroin. It may not be suitable if patients have a chronic pain syndrome – methadone provides better analgesia.
- Methadone treatment is more successful if carried out slowly or with a linear dose reduction; methadone can suppress withdrawal for 24 – 36 hours so once daily dosing is sufficient.

3. Adverse effects

- Buprenorphine is preferable to $\alpha 2$ adrenergic agonists if there are concerns about bradycardia or hypotension

4. *Withdrawal severity*

- Buprenorphine results in lower severity of withdrawal symptoms than $\alpha 2$ adrenergic agonists

The **dose** of methadone or buprenorphine depends very much on the amount of street opioid used by the client – titration according to withdrawals is necessary, and this must be individualised. A dose of 40-60mg/day of methadone reduces withdrawal symptoms. But doses greater than 60mg/day are generally required to block craving for opioids.

[Buprenorphine is a partial agonist. In a highly dependent patient, injection of buprenorphine may produce mild agonistic effects which may not be perceived due to a high degree of dependency. Higher doses will have antagonistic effects and may produce withdrawal symptoms in a dependent patient.]

Ultra rapid withdrawal by giving antagonists is not advocated routinely; tight medical supervision is necessary with facilities for general anaesthesia if required.

★ **Relapse prevention with maintenance treatment:**

- **Methadone** maintenance treatment is an appropriate treatment option for opioid-dependent patients who are unwilling to undergo withdrawal or have had repeated unsuccessful attempts at withdrawal. It is effective in reducing heroin use, injecting, and sharing injecting equipment. Methadone maintenance treatment is more effective at doses in the range 60 mg to 120 mg than at lower doses. Methadone maintenance treatment should be provided in conjunction with psychosocial interventions.
- **Buprenorphine** is an effective intervention for use in the maintenance treatment of opioid dependent patients who are unwilling to undergo withdrawal or who have had repeated unsuccessful attempts at withdrawal. It can be offered as an alternative to methadone maintenance treatment, though may not be as effective as high dose methadone maintenance treatment due to higher dropouts. Higher daily doses of buprenorphine (at least 8 mg and probably as high as 16 mg) are more effective in maintenance treatment.
- **Levo-alpha-acetylmethadol (LAAM)** is a long-acting congener of methadone that suppresses withdrawal for 48 to 72 hours and thus has the advantage of requiring less frequent clinic visits than methadone. It is no longer used as patients developed prolonged QT intervals associated with fatal torsades de pointes.
- **Suboxone** contains buprenorphine as well as the opioid antagonist naloxone to deter the abuse of tablets by intravenous injection (naloxone does not work on sublingual administration; so when IV misuse of buprenorphine is attempted, this . Controlled trials suggest that buprenorphine and naloxone at a 4:1 ratio will produce unpleasant withdrawal symptoms if taken intravenously by patients who are addicted to opioids, but has no such effect when taken sublingually. However, the Suboxone formulation still has

potential to produce an opioid agonist "high" if injected by non-dependent persons. Suboxone is approved by the European regulating body.

★ **Relapse prevention using naltrexone:**

- Naltrexone is a long-acting (72 hours) antagonist used to nullify effects of heroin if relapse occurs – thus preventing a 'lapse' becoming a relapse.
- The evidence base to support use of naltrexone for opioid relapse prevention is currently inadequate.
- Naltrexone may be a useful adjuvant in therapy, for highly motivated patients and for those who fear severe consequences if they do not stop opioid use (e.g. health care professionals)
- Naltrexone implants and depot injections are untested in Phase III randomized controlled trials and cannot be recommended at this time
- The following side-effects have been most frequently reported: with naltrexone: Insomnia, anxiety, nervousness and dysphoria, abdominal pain/cramps, nausea and vomiting, low energy, joint and muscle pain, headaches and depression. It might precipitate serious withdrawal syndrome if prescribed in an ongoing heroin user.

NICE opioid detoxification summary

For all patients who are opioid dependent and have expressed an informed choice to become abstinent, services should:

- Offer detoxification readily;
- Provide detailed information about detoxification especially the withdrawal experience, management approaches, loss of opioid tolerance on successful detoxification and so the risk of intoxication rises significantly.
- Offer a **community based detoxification** programme routinely, except when
 - Previous failure of community detoxification.
 - Significant additional physical or mental health problems
 - Polydrug detoxification
 - Considerable social problems
- Offer **buprenorphine** or **methadone** as first line treatment, with due consideration to service user's preference.
- Detoxification should normally be started with the same medication as being used for maintenance in patient.
- Consider **lofexidine**, especially for those with mild or uncertain dependence, but warn patients that this necessitates the use of adjunct medications to manage withdrawal symptoms such as nausea, vomiting and shivering
- Lofexidine detox outcomes are no better than for buprenorphine or methadone;
- **Do not routinely use** drugs such as benzodiazepines, minor analgesics, or antidiarrhoeals to manage opioid withdrawal symptoms.

Psychosocial interventions:

- Offer *opportunistic brief interventions* focused on motivation to people in limited contact with services
- Routinely provide information about *self-help groups*, often based on 12-step principles (for example, Narcotics Anonymous and Cocaine Anonymous).
- Drug misuse services
- **Contingency management** with incentives such as vouchers or privileges, contingent on each presentation of a drug-negative test result (for example, free from cocaine or non-prescribed opioids);
- Vouchers should have monetary values that start at about £2 (€3; \$4) and increase with each additional, continuous period of abstinence;

Urine drug tests: (three tests a week for the first three weeks, two tests a week for the next three weeks and once weekly thereafter until stability is achieved).

5. Treatments for other substances

- ★ **Treatment for amphetamine dependence:** Most users do not become dependent, but in some, dependence and craving on withdrawal can be very rapid. The evidence supporting pharmacological treatment is weak for stimulant use. Psychosocial treatments such as cognitive behaviour therapy and contingency management are recommended as the main treatment approaches albeit with poor quality evidence. Amineptine was tested in two RCTs with favourable results, but it has been withdrawn because of abuse potential. Harm reduction includes injection-related safety measures.
- ★ **Treatment for cocaine withdrawal:** No pharmacological agents can reduce the intensity of cocaine withdrawal or craving reliably; dopamine agonists such as *amantadine*, *bromocriptine* and *pergolide* have been investigated, but a meta-analysis of these studies found no significant difference between interventions. The evidence does not support the use of dopamine agonists or antidepressants, carbamazepine or olanzapine in treating cocaine dependence.
- ★ **Contingency management in cocaine users:**
 - These are interventions in which clients receive tangible, positive reinforcers for objective evidence of behaviour change. For example, patients receive a voucher, exchangeable for retail goods and services, whenever they submit a drug-negative urine specimen.
 - A contingency management technique termed the fish-bowl procedure involves allowing patients to draw a voucher from a fish-bowl that can be exchanged for prizes worth between \$1 and \$100, after each negative drug test.
 - Contingency management has RCT evidence supporting its use (either voucher alone or with an intensive behaviour therapy) in maintaining cocaine abstinence (Higgins et al., 2003). It is also effective (alone or in combination with cognitive therapy) in cocaine-dependent methadone-maintained opioid addicts (Rawson et al., 2002) though the effect is seen for cocaine rather than opioid abstinence.
 - The NICE guidelines recommend that contingency management should be used for the treatment of primary stimulant misuse and for illicit drug use in methadone maintenance treatment. This follows that contingency management had the most consistent evidence base for all psychosocial interventions for these groups. A major problem with contingency management is the reduction or loss of response when the reinforcement is discontinued.
- ★ **Treatment to aid smoking cessation:** The primary goal of treatment is a permanent cessation of smoking. An abstinent period of 6 months or longer is widely regarded as an acceptable outcome in trials; relapse rates after 6m of cessation are low (8% per year for the first few years reduces further). Without assistance, only 5% to 10% of smokers making a quit attempt succeed in achieving long-term tobacco abstinence.

- **Nicotine replacement therapy (NRT):** Nicotine replacement therapy increases the chance of achieving abstinence for at least 6 months [OR = 1.74, meta-analyses from 96 RCTs]. NRT is usually given for up to 2 weeks after quitting.
 - The treatment is available as gums, patches, inhalers, nasal spray, sublingual tablets and lozenges. All NRT products are equally efficacious. Though efficacy did not differ among products, but patient compliance might be different. Combinations of NRT products can safely be used (e.g. patch plus inhalator) and may be more effective. Higher dose products are more effective for heavily dependent smokers.
 - The percentage of patients who were compliant in the first week of treatment in a study - 82% for the patch, 38% for the gum, 15% for the nasal spray, and 11% for the inhaler.
 - The transdermal nicotine patch is the slowest to act but delivers a relatively constant dose of nicotine over 16 to 24 hours. The nicotine nasal spray is the fastest, with blood nicotine levels reaching a peak 5 to 10 minutes after administration. Nicotine in the gum and inhaler is absorbed through the oral/pharyngeal mucosa, and peak blood levels of nicotine are reached in 20 minutes. None of the above delivers nicotine as fast as smoking a cigarette.
 - Additional behavioural support (as provided in specialist smokers' clinics) improves overall success rates but is not required for the pharmacotherapy per se to be effective.
 - NRT can be used safely in patients with stable cardiac disease – it is safer than smoking. Although effectiveness is not established for pregnant women and for young people. (Under 18 years), it must be considered on a case-to-case basis. There is insufficient evidence to match types of the smoker to different NRT products. Higher dose patches are more effective than lower dose patches in general. Some smokers transfer their dependence on cigarettes to NRT.
- **Bupropion:** Bupropion is not licensed for adolescents or pregnant women, and is contraindicated for people with a history of seizures or eating disorders.
 - The anti-smoking effect appears to be independent of a past history of depression
 - The combination of bupropion and nicotine patch was not significantly better than bupropion alone. Bupropion is started at 150 mg each day for six days; after which it is increased to 300mg/day. Smoking must be stopped on day eight of treatment. In total eight weeks of treatment is required to quit smoking.
 - **NICE** recommends that if a smoker's attempt to quit is unsuccessful, one should not usually prescribe NRT or bupropion for another attempt within **6 months**.
- **Other drugs:**
 - **Nortriptyline** has also shown efficacy for smoking cessation in some studies.
 - Both oral and transdermal **clonidine** reduce withdrawal symptoms and have been effective in producing long-term cessation in some, but not all, trials (Gourlay and Benowitz 1995).
 - **Varenicline (Champix)** is a partial agonist at the $\alpha 4\beta 2$ unit of nicotinic acetylcholine receptor. It assists smoking cessation by relieving nicotine withdrawal symptoms and reducing the rewarding properties of nicotine.
-

6. Non-substance addictive behaviours

Pathological gambling

- ★ Originally classified as an Impulse Control Disorders alongside pyromania and trichotillomania in DSM-IV, this is now moved alongside substance use disorders in DSM-5.
- ★ There are multiple lines of evidence for overlap between pathological gambling and the substance use disorders. Pathological gamblers display irritability when attempting to reduce gambling (withdrawal), a tendency to gamble with progressively higher stakes (tolerance), with ~ 30-50% showing comorbid substance misuse.
- ★ Genetic markers influencing dopamine transmission, and personality traits linked to impulsivity are considered to be associated with the condition.
- ★ Naltrexone (esp. higher doses) has been shown to have a meaningful efficacy in reducing the gambling frequency (2 favourable RCTs). SSRIs such as fluvoxamine (1 favourable and 1 negative RCT), citalopram (1 favourable open label), and sertraline (1 negative RCT) and the mood stabilizer lithium (2 favourable RCTs) have some evidence for treating pathological gamblers. Psychological treatments seem to have high drop-out rates though CBT appears to be promising. (For a detailed review please see George & Murali et al., 2007)

Internet addiction

- ★ While not a diagnosis as per ICD or DSM, it is currently added to the index of DSM 5 for further research.
- ★ It was first described in 1998 by Young. It has been categorized into five types: cybersexual addiction; cyberrelationship addiction; net compulsion (e.g. gambling or shopping on the internet); information overload (e.g. compulsive database searching); and computer addiction (now called gaming addiction).
- ★ Appears to be more often reported in China, Taiwan and Southeast Asia than in Europe or Americas.

Compulsive Buying

- ★ Also called '**oniomania**', this is NOT a recognised psychiatric disorder in DSM or ICD. It is defined as a problematic buying behaviour that is: (1) uncontrollable; (2) markedly distressing, time-consuming, and/or resulting in family, social, vocational, and/or financial difficulties; and (3) not occurring only in the context of hypomanic or manic symptoms" (Murali et al., 2012). An estimated prevalence of compulsive buying traits has been noted in 2-8% with >80% female cases in survey samples.
- ★ A lifetime prevalence of any affective disorder of 68%, predominantly depression, has been noted in people with compulsive buying disorder.
- ★ Due to the biological and phenomenological similarities between compulsive shopping and obsessive-compulsive disorder, SSRIs have been evaluated. Open label trials support the use of

high-dose citalopram, escitalopram and other SSRIs. RCT evidence exists for the efficacy of cognitive-behavioural therapy with self-monitoring.

7. Unconventional substances

A. Misuse of anabolic steroids

- ★ Some bodybuilders and athletes use anabolic steroids as an aid to enhance performance (ergogenic agent). Anabolic steroids belong to class C (Controlled Drugs) under the Misuse of Drugs Act in the UK.
- ★ ICD-10 includes steroid as a non-psychoactive substance along with hormones and classifies steroid abuse along with disorders of physiology (F50.xx) rather than addiction (F10.xx).
- ★ Some fitness centre surveys indicate as much as 13% of users being exposed to anabolic steroids. They are considered to support the frequency and intensity of workouts, enhance muscle bulk, enhance strength and endurance, and speed up healing from sports injuries.
- ★ They are mostly taken orally or IM; some topical forms are also available
- ★ The doses consumed by drug-users are nearly 10–100 times greater than therapeutic or physiological doses.
- ★ The abuse tends to follow one of the three patterns: ‘cycling’ (using for 4-12 weeks in a cyclical fashion), ‘stacking’ (regular use of multiple preparations) and ‘pyramiding’ (gradually building the dose to a peak and then tapering).
- ★ Anabolic steroids are associated with increased risk of aggression, violence, psychosis, mania, depression and endocrine abnormalities resulting in acne (~50%), testicular atrophy and/or gynaecomastia (1/3rd of users).

B. Legal Highs

- ★ 5–10% of young people in Europe had taken a legal high in the previous year, compared to 25–50% for cannabis (Gallup Organization 2011).
- ★ Testing a psychoactive product to determine its harmful nature costs nearly £1 million and takes nearly 1 year. So it is practically impossible to keep pace with new products released in the market before applying statutory prohibitions on the basis of the harmful potential.

CATEGORY	PRODUCTS	EFFECTS
Stimulant-like legal highs (act via monoamine reuptake transporter – SERT, DAT or NAT blockade)	Primary agent: Mephedrone ('meow meow' or 4-methylmethcathinone or 4-MMC). <i>Other products:</i> 5-IAI, 2-AI, (benzylpiperazine)BZP, 6-APB Naphyrone, NBOMe-series 2C-series, Benzo fury, Ivory Wave. (BZP was initially developed as a veterinary anthelmintic (anti-parasite) drug)	Desired effects: Euphoria and a sense of disinhibition. Adverse effects: Serotonin syndrome, psychosis, agitation, mania, cardiovascular symptoms, hyperthermia
Psychedelic-like legal highs (act via 5HT1A, 2A and 2C receptors)	DMT (dimethyltryptamine: derivative of the essential amino acid tryptophan, traditionally extracted from a mixture of South American plant ayahuasca), Bromo DragonFLY, 5-MeO-Dalt, NBOMe-series, 2C-series, Salvia (Salvinorin A - an extract from the sage plant: D2 receptor agonist).	Desired effects: psychedelic effects such as perceptual alterations and feelings of 'nirvana' or boundlessness
Cannabis-like legal highs (act via CB1 and CB2 receptors)	JWH-018 JWH-081 JWH-122 AM-2201 UR-144 XLR-11. These products are variously packed and sold as ' <u>Spice</u> .'	Desired effects: Relaxation, pain and anxiety reduction, sedation and euphoria Adverse effects: Paranoia, psychosis, intense anxiety, rarely seizures
Benzomimetic legal-highs (GABA facilitating effects)	Pyrazolam, Flubromazepam, Phenazepam	Desired effects: Relaxation, anxiety reduction, sedation Adverse effects: Respiratory depression and withdrawal seizures
Dissociative anaesthetic like legal-highs (NMDA antagonism)	Mexxy, MXE (methoxetamine – a ketamine analogue), Diphenidine, 3 or 4-MeO-PCP, N-EK, 2-MK,	Desired effects: Dissociative state, pain reduction, weightlessness Adverse effects: Headache, psychotic symptoms, GI disturbances (nausea), cognitive impairment

Adapted from Baumeister, D., Tojo, LM and Tracy DK. Ther Adv Psychopharmacol 2015, Vol. 5(2) 97–132

8. Pregnancy and substance use

A. Alcohol Use in Pregnancy:

- Foetal alcohol syndrome is the commonest non-genetic cause of learning disability. Intellectual impairment, dysmorphic facial features and disruptive behaviour are noted.
- Women should be advised not to drink alcohol or at maximum, one drink/day
- Psychosocial interventions should be offered and be the mainstay of treatment
- Patients with symptomatic withdrawal should be offered medical cover for their detoxification, ideally as an inpatient
- Medication to sustain abstinence such as disulfiram, acamprosate or naltrexone should be avoided

B. Opioids in pregnancy:

- Major risk is the harm caused to the baby by injections, e.g., infection and the effects of drug-induced stillbirth, premature birth, antenatal complications (OR for antepartum hemorrhage is 2.33 in opiate using mothers), low birth weight (mean reduction of ~500g in heroin/methadone users), microcephaly and neonatal withdrawal.
- Ideally, the mother should be encouraged to become abstinent prior to delivery – often this may not be possible.
- Stabilization on an oral substitute drug should always be the initial aim of treating a pregnant opioid user.
- If pregnancy begins when a woman is taking high doses of methadone, the dosage should be reduced slowly (e.g., 1 mg every 3 days), and foetal monitoring must be carried out closely. There is no clear dose-dependent relationship between maternal methadone and the intensity of neonatal withdrawal, but lower doses (15 mg) are advocated during maintenance. Methadone is more likely than heroin to induce withdrawal syndrome (60-80%) in the newborn. Methadone is not a contraindication to breast-feeding but local policies may vary.
- Fewer data are available for buprenorphine maintenance, but it appears similar benefits are seen for mother and foetus as for methadone

Principles of managing pregnant opioid user:

1. Therapeutic alliance
2. Aim to reduce risk-taking behaviours (sharing needles, prostitution).
3. Stabilize on non-injectable alternatives such as methadone.
4. If considering detoxification, this should be done in the middle trimester – avoid first trimester for the fear of abortion and last trimester for the fear of premature birth.
5. Close liaison with obstetric, midwifery, and paediatric teams, and with social services where appropriate.
6. HIV and hepatitis screening (vaccination where appropriate).

C. Other drugs in pregnancy:

- ★ In pregnant women, cocaine produces **small for gestational age** babies with **microcephaly**.
- ★ Cannabis use both during pregnancy and lactation adversely affects neurodevelopment, thus influencing the future adult productivity and lifetime outcomes. There is a 2-fold risk of low birth weight, with mean weights being reduced by ~275g in the newborn.

D. Neonatal withdrawal syndromes:

Severity depends on the nature of drugs and the frequency of use by the mother during pregnancy. The table below is based on data from <http://emedicine.medscape.com/article/978492-clinical.htm>

Substance	Effect
Alcohol withdrawal:	Seen within 3-12 hours after delivery. Hyperactivity, crying, irritability, poor sucking, tremors, seizures, poor sleeping patterns, hyperphagia, and diaphoresis. Signs usually appear at birth and features of Foetal Alcohol Syndrome may be apparent later on further development.
Barbiturate withdrawal	Irritability, severe tremors, hyperacusis, excessive crying, vasomotor instability, diarrhea, restlessness, increased tone, hyperphagia, vomiting, and disturbed sleep.
Marijuana withdrawal	A mild opiate-like withdrawal syndrome with fine tremors, hyperacusis, and a prominent Moro reflex; mild and very rarely require treatment.
Nicotine withdrawal	Mild signs such as fine tremors and variations in tone; recent data indicates subtle neonatal behaviours, such as poor self-regulation and an increased need for handling.
Opiate withdrawal	Usually begins 24-48 hours after birth, depending on the time of last dose. However, signs may not appear in the infant until 3-4 days after birth. Hyperirritability, gastrointestinal dysfunction, respiratory distress, and vague autonomic symptoms (e.g., yawning, sneezing, mottling, fever). Tremors and jittery movements, high-pitched cries, increased muscle tone, and irritability are common. Normal reflexes may be exaggerated. Loose stools are common, leading to possible electrolyte imbalances and diaper dermatitis. Methadone withdrawal symptoms typically appear within 48-72 hours but may not start until the infant is aged 3 weeks. Milder with buprenorphine withdrawal.
Amphetamines and cocaine	Whether stimulants use causes, neonatal withdrawal remains unclear. Increasing evidence suggests that it is associated with an increased risk of prematurity and intrauterine growth restriction during pregnancy.
Antidepressants	Infants have been noted to show jitteriness, respiratory distress, and other neonatal complications. They are more commonly observed in newborns whose mothers were taking a short-acting SSRI such as paroxetine and less with sertraline.

DISCLAIMER: This material is developed from various revision notes assembled while preparing for MRCPsych exams. The content is periodically updated with excerpts from various published sources including peer-reviewed journals, websites, patient information leaflets and books. These sources are cited and acknowledged wherever possible; due to the structure of this material, acknowledgements have not been possible for every passage/fact that is common knowledge in psychiatry. We do not check the accuracy of drug-related information using external sources; no part of these notes should be used as prescribing information

Notes prepared using excerpts from

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